

Category	Item	Value	Unit
Material	Concrete	100	m³
	Steel	50	kg
	Brick	200	kg
	Wood	30	m³
Labor	Construction Worker	10	hr
	Electrician	5	hr
	Plumber	3	hr
	Painter	2	hr
Equipment	Excavator	1	hr
	Truck	2	hr
	Generator	1	hr
	Drill	1	hr

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9702/34

October/November 2009

2 hours

Additional Materials: As listed in the Confidential Instructions.

All questions in this paper carry equal marks.

For Examiner's Use	
1	
2	
Total	



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You may not need to use all of the materials provided.

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- 1 In this experiment, you will investigate changes in potential difference in a circuit as one of the resistances is varied.

(a) Assemble the circuit of Fig. 1.1.

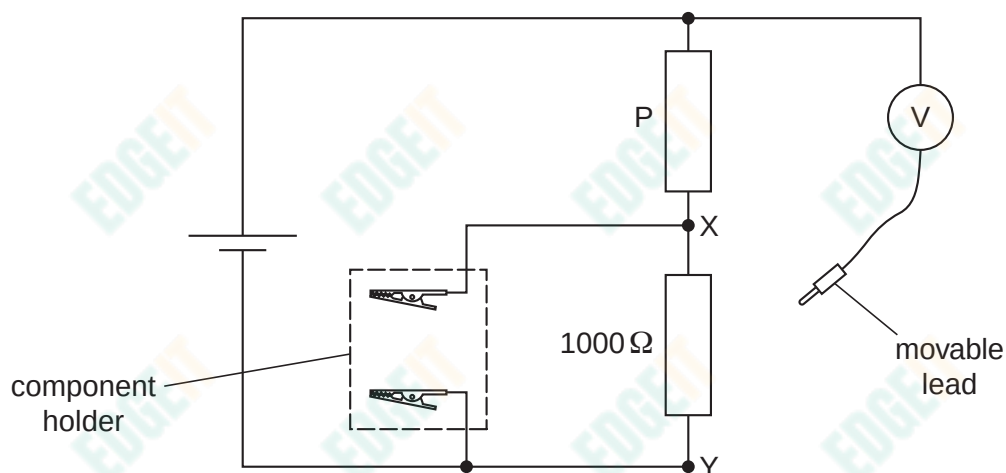


Fig. 1.1

- (b) Connect the movable lead to terminal Y and record the voltmeter reading V_0 .

$V_0 = \dots\dots\dots$

- (c) (i) Connect the movable lead to terminal X and **leave it connected there for the rest of the experiment.**

- (ii) Select one of the other labelled resistors provided and connect it between the crocodile clips on the component holder. Record the resistance R of your selected resistor and the voltmeter reading V .

$R = \dots\dots\dots \Omega$

$V = \dots\dots\dots$

- (d) Repeat (c)(ii) using a different labelled resistor until you have six sets of readings of R (measured in ohms) and V .

Disconnect the cell when you have finished your measurements.

In your table of results include columns for the values of $\frac{1}{V}$ and the values of $\frac{R}{(1000 + R)}$.

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- (e) (i) Plot a graph of $\frac{1}{V}$ on the y -axis against $\frac{R}{(1000 + R)}$ on the x -axis.
 (ii) Draw the line of best fit.
 (iii) Determine the gradient and y -intercept of the line of best fit.

gradient =

y -intercept =

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- (f) The quantities V and R are related by the equation

$$\frac{1}{V} = \frac{1000k}{P} \left(\frac{R}{1000 + R} \right) + k$$

where k is a constant and P is the resistance of resistor P .

Using your answers from (e)(iii), determine the value of P .

Give an appropriate unit.

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$P =$

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Please turn over for Question 2.

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- 2 In this experiment, you will investigate how the length of a thread of cotton affects the amount of energy it can absorb before breaking.

- (a) Use the boss to clamp the nail to the retort stand at a height of about 50 cm above the bench. Place the apparatus in the tray.
- (b) (i) Cut a 45 cm length of thread.
- (ii) Tie the ends of the thread together to make a large loop. Suspend the loop from the nail and slot the 50 g mass into it, securing the mass with sticky tape, as shown in Fig. 2.1.

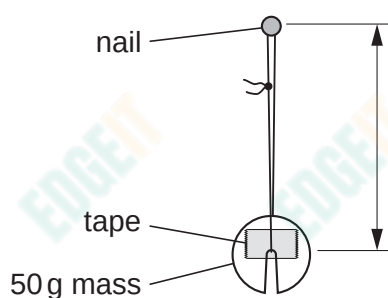


Fig. 2.1

- (c) (i) Measure the length l as shown in Fig. 2.1.

$l =$

- (ii) Lift the 50 g mass vertically a distance h of 5 cm, as shown in Fig. 2.2. Release the mass.

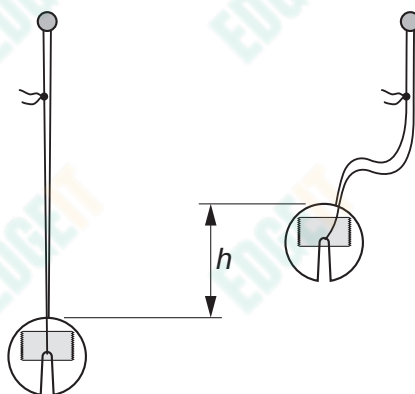


Fig. 2.2

- (iii) Keep repeating (c)(ii), each time increasing the value of h , until the thread breaks. Record the value of h at which the thread breaks.

$h_{\text{final}} =$ cm

- (d) Estimate the percentage uncertainty in your value of h_{final} .

percentage uncertainty =

- (e) Calculate the gravitational potential energy E_p given to the 50 g mass when it was lifted the distance h_{final} .

($E_p = mgh_{\text{final}}$ where $g = 9.81 \text{ m s}^{-2}$).

E_p J

- (f) Cut an 85 cm length of thread. Repeat (b)(ii), (c) and (e).

$l =$

$h_{\text{final}} =$ cm

$E_p =$ J

- (g) It is suggested that E_p and l are related by the equation

$$E_p = k\sqrt{l}$$

where k is a constant.

By calculating values of k , explain whether your results support this relationship.

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- (h) (i) State and explain four sources of error or limitations of the procedure in this experiment.

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- (ii) Suggest and explain four improvements that could be made to this experiment. You may suggest the use of other apparatus or different procedures.

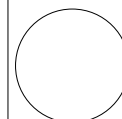
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